STEP1: Shock Propagation Along a Single Branch Unifying the Second Version's Event Definitions with the First Version's Symmetric Formulas

1 Purpose and Unified Conventions

This document retains two complementary forms of representation:

1. Use explicit event definitions to distinguish "when a production node itself stops/recovers" from "when a specific downstream node actually observes that shock";
2. For a single shutdown window, preserve the highly symmetric stop/recovery recursions and cumulative formulas of the original version;
3. For multiple windows and general DAGs, use the cumulative deficit function $D(t)$ as a unified generalization of those formulas.

The central principle is

$$\boxed{\text{Define physical events first, derive single-window formulas, then generalize to } D(t).}$$

Consider a direct production-relation branch

$$P_p \longrightarrow M_1 \longrightarrow \cdots \longrightarrow M_K \longrightarrow P_q.$$

2 The Two Types of Events Must Be Distinguished

2.1 The Production Node's Own Production-Start Event

Let $u_p(t)$ denote the actual intensity of production activity initiated by node P_p at time t, with initialized normal intensity

$$u_p^0 \equiv u_p(0).$$

If P_p stops initiating new production during a single complete-shutdown window, define

$$\boxed{I_p^{act} = [\sigma_p, \rho_p),}$$

where

$$\boxed{\sigma_p = \text{time when } P_p \text{ itself stops initiating new production,}}$$

$$\boxed{\rho_p = \text{time when } P_p \text{ itself resumes initiating production at normal intensity.}}$$

Define the duration of this internal shutdown window as

$$\boxed{\kappa_p = \rho_p - \sigma_p.}$$

Here σ_p, ρ_p describe only events on

$$\boxed{\text{the production-start plane of } P_p}$$

and do not imply that any particular downstream node has already experienced a supply interruption.

2.2 Stop/Recovery Observed by a Specific Downstream Node

For a specific direct downstream node P_q, define

$$\boxed{\begin{array}{c} t^{\text{stop}}_{p\to q} = \\ \text{the first time } P_q \text{ cannot obtain normally required inputs} \\ \text{because of the supply deficit from } P_p \end{array}}$$

and

$$\boxed{\begin{array}{c} t^{\text{rec}}_{p\to q} = \\ \text{the first time, after an effective interruption, that } P_q \\ \text{again receives normal supply from } P_p\text{'s resumed production} \end{array}}$$

Therefore, we must distinguish

$$\boxed{(\sigma_p, \rho_p) \neq (t^{\text{stop}}_{p\to q}, t^{\text{rec}}_{p\to q}).}$$

The former is a

node-local event,

The latter is a

downstream-visible event.

If P_p connects to multiple downstream nodes, then generally

$$t^{\text{stop}}_{p\to q_1} \neq t^{\text{stop}}_{p\to q_2}, \qquad t^{\text{rec}}_{p\to q_1} \neq t^{\text{rec}}_{p\to q_2},$$

because different output branches may have different delays and inventory buffers.

3 Two Basic Propagation Quantities

For a direct branch $P_p \to P_q$, let the transportation-edge and inventory-node sets be

$$\mathcal{E}_{p\to q}, \qquad \mathcal{M}_{p\to q}.$$

3.1 Total Time Shift

Define

$$\boxed{\Theta_{p\to q} = \tau^{prod}_p + \sum_{e\in\mathcal{E}_{p\to q}} \tau^{trans}_e.} \tag{1}$$

Here, the upstream process P_p's own production delay is explicitly included in the downstream propagation from P_p to P_q.

Therefore $\Theta_{p\to q}$ represents only

$$\boxed{\text{time shift}}$$

and does not change the duration of a deficit.

3.2 Total Inventory Coverage

For each inventory node $m \in \mathcal{M}_{p\to q}$, denote its initialized normal outflow rate by

$$f_m^{out,0} > 0.$$

Define inventory coverage

$$c_m = \frac{q_m(0)}{f_m^{out,0}},$$

The total inventory coverage time of the entire branch is then

$$\boxed{C_{p\to q} = \sum_{m\in\mathcal{M}_{p\to q}} \frac{q_m(0)}{f_m^{out,0}}.} \tag{2}$$

Physically:

Delay controls shock arrival; inventory controls remaining deficit duration.

4 Main STEP1 Result: Symmetric Propagation Formulas for a Single Shutdown Window

Let P_p's own production-start gap be

$$[\sigma_p, \rho_p), \qquad \kappa_p = \rho_p - \sigma_p.$$

Define the candidate supply-interruption front

$$\boxed{s_{p\to q} = \sigma_p + \Theta_{p\to q} + C_{p\to q},} \tag{3}$$

and the arrival front of resumed production output

$$\boxed{r_{p\to q} = \rho_p + \Theta_{p\to q}.} \tag{4}$$

If

$$s_{p\to q} < r_{p\to q},$$

equivalently

$$\boxed{\kappa_p > C_{p\to q},} \tag{5}$$

then this internal shutdown of P_p penetrates all inventories on the branch and affects P_q.

In this case, the central STEP1 formulas are

$$\boxed{\begin{aligned} t_{p\to q}^{\text{stop}} &= \sigma_p + \Theta_{p\to q} + C_{p\to q}, \\ t_{p\to q}^{\text{rec}} &= \rho_p + \Theta_{p\to q}. \end{aligned}} \tag{6}$$

This preserves the most direct symmetric structure of the original version:

$$\boxed{\begin{aligned} \text{stop} &= \text{upstream stop} + \text{delay} + \text{inventory coverage}, \\ \text{recovery} &= \text{upstream recovery} + \text{delay}. \end{aligned}} \tag{7}$$

Subtracting the two formulas yields

$$\boxed{t_{p\to q}^{\text{rec}} - t_{p\to q}^{\text{stop}} = (\rho_p - \sigma_p) - C_{p\to q}.} \tag{8}$$

Thus the effective duration of the supply interruption is

$$\boxed{\Delta_{p\to q} = \left[(\rho_p - \sigma_p) - C_{p\to q}\right]_+ .} \tag{9}$$

If

$$s_{p\to q} \geq r_{p\to q},$$

then

$$\boxed{\Delta_{p\to q} = 0.} \tag{10}$$

In this case, although P_p itself did indeed shut down, the resumed-output front catches up to the deficit front before the inventories are exhausted, so P_q never experiences a genuine effective supply interruption. Here $r_{p\to q}$ can be called the "resumed-output arrival front," but it should not be described as P_q "shutting down and then recovering."

5 Standard $P - M - M - P$ Structure

Consider

$$P_p \to M_1 \to M_2 \to P_q.$$

Define

$$\boxed{\begin{aligned} \Theta_{p\to q} &= \tau_p^{prod} + \tau_{P_p\to M_1}^{trans} + \tau_{M_1\to M_2}^{trans} + \tau_{M_2\to P_q}^{trans}, \\ C_{p\to q} &= \frac{q_{M_1}(0)}{f^0_{M_1\to M_2}} + \frac{q_{M_2}(0)}{f^0_{M_2\to P_q}}. \end{aligned}} \tag{11}$$

If a single shutdown window penetrates all inventories, then

$$\boxed{\begin{aligned} t^{\text{stop}}_{p\to q} &= \sigma_p + \tau_p^{prod} + \tau_{P_p\to M_1}^{trans} + \frac{q_{M_1}(0)}{f^0_{M_1\to M_2}} \\ &\quad + \tau_{M_1\to M_2}^{trans} + \frac{q_{M_2}(0)}{f^0_{M_2\to P_q}} + \tau_{M_2\to P_q}^{trans}, \\ t^{\text{rec}}_{p\to q} &= \rho_p + \tau_p^{prod} + \tau_{P_p\to M_1}^{trans} + \tau_{M_1\to M_2}^{trans} + \tau_{M_2\to P_q}^{trans}. \end{aligned}} \tag{12}$$

Thus, production and transportation delays appear perfectly symmetrically in the stop and recovery fronts; inventory coverage enters only the stop front.

6 Single-Source Linear Chain: Recovering the Original Recursive and Cumulative Forms

Consider

$$P_0 \to \mathcal{B}_1 \to P_1 \to \mathcal{B}_2 \to P_2 \to \cdots \to \mathcal{B}_i \to P_i,$$

where $\mathcal{B}_j$ denotes a material branch from P_{j-1} to P_j containing no other production nodes.

Define

$$\Theta_j \equiv \Theta_{P_{j-1}\to P_j}, \qquad C_j \equiv C_{P_{j-1}\to P_j}.$$

Suppose the source P_0 experiences a single production-start gap

$$[\sigma_0, \rho_0).$$

In a single-input linear chain, as long as the shock has not been fully absorbed by inventory, the downstream-visible shortage from P_{j-1} to P_j directly determines P_j's own production-start gap. Hence the recursion is

$$\boxed{\begin{aligned}\sigma_j &= \sigma_{j-1} + \Theta_j + C_j,\\ \rho_j &= \rho_{j-1} + \Theta_j.\end{aligned}} \tag{13}$$

Summing segment by segment gives

$$\boxed{\begin{aligned}\sigma_i &= \sigma_0 + \sum_{j=1}^{i} \Theta_j + \sum_{j=1}^{i} C_j,\\ \rho_i &= \rho_0 + \sum_{j=1}^{i} \Theta_j.\end{aligned}} \tag{14}$$

Therefore

$$\boxed{\rho_i - \sigma_i = (\rho_0 - \sigma_0) - \sum_{j=1}^{i} C_j.} \tag{15}$$

If the earlier version assumes

$$\sigma_0 = 0, \qquad \rho_0 = k,$$

we immediately obtain

$$\boxed{\begin{aligned}\sigma_i &= \sum_{j=1}^{i} (\Theta_j + C_j),\\ \rho_i &= k + \sum_{j=1}^{i} \Theta_j,\end{aligned}} \tag{16}$$

and

$$\boxed{\rho_i - \sigma_i = k - \sum_{j=1}^{i} C_j.} \tag{17}$$

If

$$k \le \sum_{j=1}^{i} C_j,$$

then the shock has already been fully absorbed by accumulated inventory coverage before reaching P_i, so P_i develops no effective production-start gap.

7 From the Original Cumulative Formulas to the Cumulative Deficit Function $D(t)$

The original symmetric formulas are suitable for describing

$$\boxed{\text{a single stop/recovery window.}}$$

To handle multiple disjoint windows, branching after merging, and general DAGs, define the cumulative internal production-start deficit of P_p as

$$\boxed{D_p^{act}(t) = \text{Equivalent normal-production time not initiated by } P_p \text{ up to time } t.} \tag{18}$$

Under the complete-shutdown / complete-recovery baseline,

$$\dot{D}_p^{act}(t) \in \{0, 1\}$$

holds almost everywhere, and

$$\boxed{u_p(t) = u_p^0 \left[1 - \dot{D}_p^{act}(t)\right].} \tag{19}$$

For a single internal shutdown window

$$[\sigma_p, \rho_p),$$

we have

$$\boxed{D_p^{act}(t) = \min\left\{(t - \sigma_p)_+, \rho_p - \sigma_p\right\}.} \tag{20}$$

Adopt the convention that when $s < 0$,

$$D_p^{act}(s) = 0.$$

Then the STEP1 propagation operator for arbitrary windows is defined as

$$\boxed{D_{p\to q}(t) = \mathcal{T}_{p\to q}[D_p^{act}](t) = \left[D_p^{act}(t - \Theta_{p\to q}) - C_{p\to q}\right]_+.} \tag{21}$$

Here

$$(x)_+ \equiv \max\{x, 0\}.$$

Its basic components are

$$\boxed{\text{Delay} : D(t) \mapsto D(t - \tau),} \tag{22}$$

and

$$\boxed{\text{Inventory} : D(t) \mapsto \left[D(t) - \frac{q(0)}{f^0}\right]_+.} \tag{23}$$

Since

$$[[x - c_1]_+ - c_2]_+ = [x - c_1 - c_2]_+,$$

the coverage times of multiple sequential inventories can be added directly.

8 How the $D(t)$ Operator Reduces to the Original Symmetric Formulas

Substitute the single-window equation (20) into (21):

$$D_{p\to q}(t) = \left[\min\left\{(t - \Theta_{p\to q} - \sigma_p)_+, \rho_p - \sigma_p\right\} - C_{p\to q}\right]_+. \tag{24}$$

If

$$\rho_p - \sigma_p > C_{p\to q},$$

then the time when $D_{p\to q}(t)$ begins to increase is precisely

$$\boxed{t_{p\to q}^{\text{stop}} = \sigma_p + \Theta_{p\to q} + C_{p\to q},}$$

and the time when it stops increasing is precisely

$$\boxed{t_{p\to q}^{\text{rec}} = \rho_p + \Theta_{p\to q}.}$$

Therefore

$$\boxed{\begin{array}{c} D(t) \text{ is not a separate model;} \\ \text{it generalizes the original stop/recovery formulas to multiple windows.} \end{array}} \tag{25}$$

9 Operator Composition and the Original "Segment-by-Segment Summation"

Define

$$\mathcal{T}_{\Theta,C}[D](t) = [D(t-\Theta) - C]_+.$$

Across two branch segments:

$$D_1(t) = \mathcal{T}_{\Theta_1,C_1}[D_0](t),$$

$$D_2(t) = \mathcal{T}_{\Theta_2,C_2}[D_1](t).$$

Using

$$[[x - C_1]_+ - C_2]_+ = [x - C_1 - C_2]_+,$$

we obtain

$$\boxed{D_2(t) = [D_0(t - \Theta_1 - \Theta_2) - (C_1 + C_2)]_+ .} \tag{26}$$

Therefore, repeated composition along a single-source linear chain gives

$$\boxed{D_i^{act}(t) = \left[D_0^{act}\left(t - \sum_{j=1}^{i} \Theta_j \right) - \sum_{j=1}^{i} C_j \right]_+ .} \tag{27}$$

Thus, on a chain without merges, the new operator agrees exactly with the original rule of summing delays and inventory coverage segment by segment.

10 Correspondence with Daily Production Status for Plant i on Day k

$D_i^{act}(t)$ is the continuous-time cumulative representation of the original daily production states.

Under the binary baseline,

$$\boxed{\frac{u_i(t)}{u_i^0} = 1 - \dot{D}_i^{act}(t).} \tag{28}$$

Therefore

$$\dot{D}_i^{act}(t) = 0 \quad \Longleftrightarrow \quad P_i \text{ produces normally,}$$

$$\dot{D}_i^{act}(t) = 1 \quad \Longleftrightarrow \quad P_i \text{ is completely shut down.}$$

If day k is defined as

$$[k, k+1),$$

and no additional intraday switching takes place, we can write

$$\boxed{x_i[k] = 1 - \left(D_i^{act}(k+1) - D_i^{act}(k) \right),} \tag{29}$$

where

$$x_i[k] \in \{0, 1\}.$$

Then

$$\boxed{x_i[k] = 1 \Longleftrightarrow \text{Plant } i \text{ operates on day } k,}$$

$$\boxed{x_i[k] = 0 \Longleftrightarrow \text{Plant } i \text{ is shut down on day } k.}$$

If stop/recovery switching is allowed within a day, then

$$D_i^{act}(k+1) - D_i^{act}(k) \in [0, 1]$$

naturally represents the proportion of normal production time lost on that day.

11 Defining Stop/Recovery for Multiple Windows

If P_p experiences multiple disjoint production-start gaps, the same downstream inventory must not be reinitialized to $q(0)$ for each window.

Thus one cannot apply

$$C_{p\to q}$$

separately to each window and simply take the union of the resulting intervals.

The correct procedure is first to calculate the entire

$$D_{p\to q}(t) = [D_p^{act}(t - \Theta_{p\to q}) - C_{p\to q}]_+.$$

Under the complete-shutdown / complete-recovery baseline, all maximal connected intervals satisfying

$$\dot{D}_{p\to q}(t) = 1$$

are defined as

$$\boxed{[t_{p\to q}^{\mathrm{stop},\ell}, t_{p\to q}^{\mathrm{rec},\ell}), \qquad \ell = 1, \ldots, L_{p\to q}.} \tag{30}$$

Thus, in the general case,

$$\boxed{D_{p\to q}(t) \text{ is the primary interface,} \qquad (t^{\mathrm{stop}}, t^{\mathrm{rec}}) \text{ is an event label read from } D.}$$

12 Normal Flow Rates and Fixed Proportions

If the output coefficient of P_p for some product Y is

$$b_{pY} > 0,$$

and a fixed fraction

$$\rho_{p\to q} > 0$$

is allocated to the branch leading to P_q, with nonbinding output-edge capacity, then the initialized normal flow rate on that branch is

$$\boxed{f_{p\to q}^0 = \rho_{p\to q} b_{pY} u_p^0.} \tag{31}$$

Fixed output coefficients and fixed allocation ratios only rescale material quantities. After normalizing deficits by their corresponding normal rates into "equivalent normal-production time," they do not alter the shape of $D(t)$; they enter inventory coverage through the normal flow rates

$$q(0)/f^0.$$

13 Applicability Limits of the Current Rules

The foregoing closed-form expressions and operators rely on the following baseline:

- The network is initialized at steady state for $t < 0$;
- Production and transportation delays are fixed;
- Production recipes and output allocation ratios are fixed;
- Output-edge capacity is nonbinding;
- No active restocking, overproduction, or rerouting;

- When normal flow resumes into a depleted inventory, it may pass through immediately without first replenishing the stock to $q_m(0)$;
- No losses, scrapping, or shelf-life constraints;
- The main focus is binary production intensity: complete shutdown or complete recovery.

If partial recovery is allowed later, the cumulative deficit function still applies, but generally

$$0 \le \dot{D}(t) \le 1,$$

and

$$u_p(t) = u_p^0[1 - \dot{D}_p^{act}(t)]$$

can describe partial production.

If an equipment failure directly freezes WIP already inside the process, production delay is no longer a simple pure delay, and an explicit WIP state is required; this case is not considered here.

14 Final STEP1 Interface

For a single window, use the intuitive symmetric formulas first

$$\boxed{\begin{aligned} t_{p\to q}^{\text{stop}} &= \sigma_p + \Theta_{p\to q} + C_{p\to q}, \\ t_{p\to q}^{\text{rec}} &= \rho_p + \Theta_{p\to q}, \end{aligned}}$$

and check

$$\rho_p - \sigma_p > C_{p\to q}.$$

For multiple windows or general DAGs, use the unified operator

$$\boxed{D_{p\to q}(t) = \left[D_p^{act}(t - \Theta_{p\to q}) - C_{p\to q}\right]_+ .}$$

STEP1 can therefore be summarized as

$$\boxed{\underbrace{D_p^{act}}_{\text{upstream node's own activity}} \xrightarrow{\substack{\text{production/transport delay} \\ \text{inventory absorption}}} \underbrace{D_{p\to q}}_{\text{shortage visible to a specific downstream node}} .}$$

This $D_{p\to q}(t)$ is exactly the input received directly by STEP2 during its required-input merge.